

THE FATE OF THE UNIVERSE

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ALTHOUGH IT IS POSSIBLE THAT THE UNIVERSE will last forever, the types of structures that exist in it today, such as planets, stars, and galaxies, almost certainly will not. At some distant point in the future, our galaxy and others will either be ripped apart; suffer a protracted, cold death; or, in the least likely scenario, be crushed out of existence in a reverse of

the Big Bang. Which of these fates befalls the universe depends to a considerable extent on the nature of dark energy—a mysterious, gravity-opposing force recently found to be playing a major part in the universe’s large-scale behavior.

BIG CRUNCH AND BIG CHILL

Until recently, cosmologists assumed that the universe’s expansion rate (see pp.44–45) must be slowing due to the “braking” effects of gravity. They also believed that a single factor—the universe’s mass-energy density—would decide which of two basic fates awaited it. Cosmologists measure the density of both mass and energy together since Einstein demonstrated that mass and energy are equivalent and interchangeable (see p.41). They calculated that if this density was above a critical value, gravity would eventually cause the universe to stop expanding and collapse in a fiery, all-annihilating implosion (a “Big Crunch”). If, however, the universe’s density was below or exactly on the critical value, the universe would expand forever, albeit with its expansion rate gradually slowed by gravity. In this case, the universe would end in a lengthy, cold death (a “Big Chill”). Research aimed at resolving this issue found that the universe has properties suggesting that it is extremely close to being “flat” (opposite), with a density of exactly the critical value. Even though some of the mass-energy in the universe needed to render it flat seemed hard to locate, its density must be near the critical value, and so its most likely fate was eternal expansion. However, in the late 1990s, models of the fate of the universe were thrown into confusion by new findings indicating that the universe’s expansion is not slowing down at all.

FOUR POTENTIAL FATES

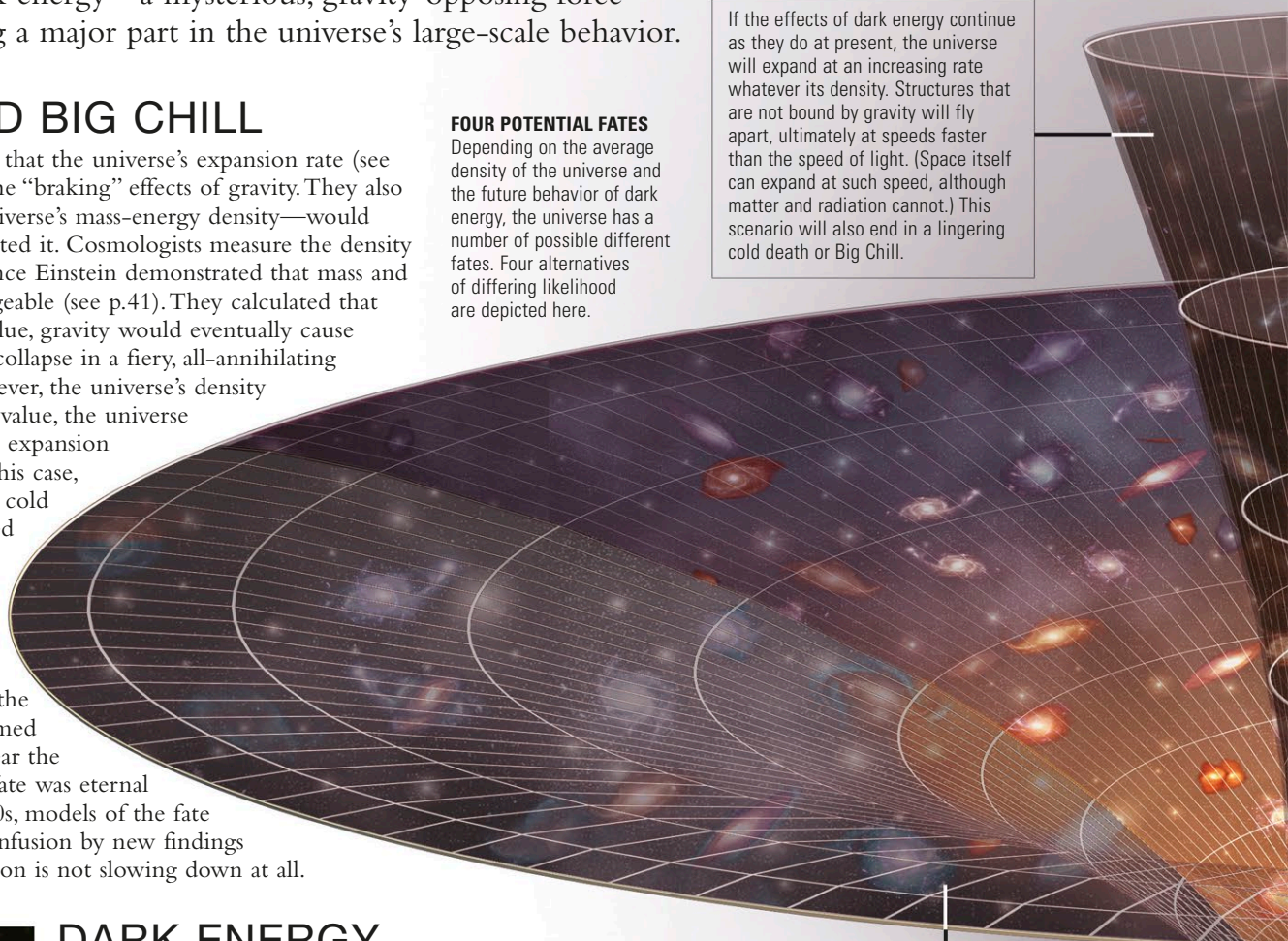
Depending on the average density of the universe and the future behavior of dark energy, the universe has a number of possible different fates. Four alternatives of differing likelihood are depicted here.

BIG CHILL

If the universe has a mass-energy density close to or just less than the critical value, and should the effects of dark energy tail off, the universe might continue to expand at a rate that slowly decreases but never comes to a complete halt. Over unimaginably long periods of time, it suffers a lingering cold death or “Big Chill.”

MODIFIED BIG CHILL

If the effects of dark energy continue as they do at present, the universe will expand at an increasing rate whatever its density. Structures that are not bound by gravity will fly apart, ultimately at speeds faster than the speed of light. (Space itself can expand at such speed, although matter and radiation cannot.) This scenario will also end in a lingering cold death or Big Chill.

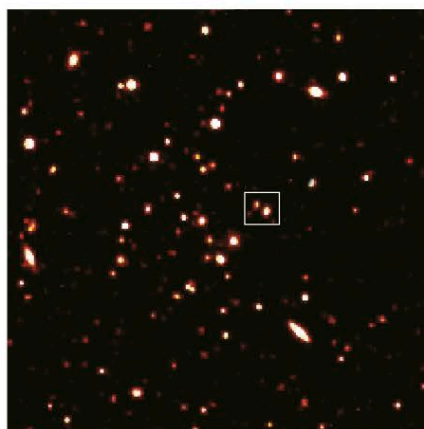


DARK ENERGY

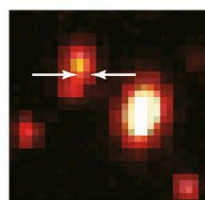
The new findings (see above) came from studies of supernovae in remote galaxies. The apparent brightness of these exploding stars can be used to calculate their distance, and by comparing their distances with the red shifts of their home galaxies, scientists can calculate how fast the universe was expanding at different times in its history. The calculations showed that the expansion of the universe is accelerating and that some repulsive force is opposing gravity, causing matter to fly apart. This force has been called dark energy, and its exact nature is uncertain, though it appears similar to a gravity-opposing force, the “cosmological constant,” proposed by Albert Einstein as part of his theory of general relativity (see pp.42–43). The existence of dark energy also accounts for the missing mass-energy in the universe required to make it flat (above) and modifies the number of possible fates for the universe.

SUPERNOVAE CLUES

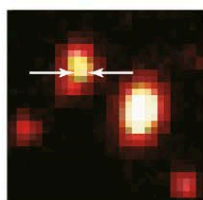
Type Ia supernovae, like that depicted here, all have the same intrinsic brightness. Thus, their apparent brightness reveals their distance.



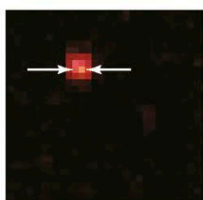
SUPERNOVA DISCOVERY



3 WEEKS BEFORE



AFTER SUPERNOVA



DIFFERENCE

BIG RIP

If the strength of dark energy increased, it could overcome all the fundamental forces and totally disintegrate the universe in a “Big Rip.” This could happen 20–30 billion years from now. First, galaxies would be torn apart, then solar systems. A few months later, stars and planets would explode, followed shortly by atoms. Time would then stop.

DARK ENERGY DOMINANCE

Dark energy provides 68 percent of the mass-energy density of the universe. Atom-based matter (in stars and the interstellar medium) and neutrinos contribute just 4.9 percent.

MASS-ENERGY CONTRIBUTIONS

